With $\frac{1}{\tau} \neq 0$: Faraday rotation + cyclotron resonance.

$$\frac{d\vec{v}}{dt} = \frac{q}{m}\vec{E}(\omega)e^{-i\omega t} + \frac{q}{m}\vec{v}\times\vec{B}_0 - \frac{\vec{v}}{\tau}$$

↳ static external field

$$\vec{v}\times\vec{B}_0 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ 0 & 0 & B_0 \end{vmatrix} = \hat{x}\, v_y B_0 - \hat{y}\, v_x B_0$$

let us split $\vec{v} = \vec{v}_\perp + \vec{v}_z$

$$\vec{E} = \vec{E}_\perp + \vec{E}_z$$

And $\vec{v}(t) = \vec{v}(\omega)e^{-i\omega t}$

$$\left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp = \frac{q}{m}\vec{E}_\perp(\omega) + \frac{q}{m}\vec{v}_\perp\times\vec{B}_0$$

$$\left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp\times\vec{B}_0 = \frac{q}{m}\vec{E}_\perp\times\vec{B}_0 + \frac{q}{m}\left(\vec{v}_\perp\times\vec{B}_0\right)\times\vec{B}_0$$

$$-\vec{B}_0\times\left(\vec{v}_\perp\times\vec{B}_0\right)$$

$$= (-)\left(\vec{v}_\perp B_0^2 - \vec{B}_0\underbrace{\left(\vec{v}_\perp\cdot\vec{B}_0\right)}_{=0}\right)$$

$$\Rightarrow \left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp\times\vec{B}_0 = \frac{q}{m}\vec{E}_\perp\times B_0 - \frac{qB_0^2}{m}\vec{v}_\perp$$

$$\Rightarrow \left(-i\omega + \frac{1}{\tau}\right)\vec{v}_\perp = \frac{q}{m}\vec{E}_\perp(\omega) + \frac{q}{m}\,\frac{\frac{q}{m}\vec{E}_\perp(\omega)\times\vec{B}_0 - \frac{q}{m}B_0^2\vec{v}_\perp}{-i\omega + \frac{1}{\tau}}$$

$$\left\{\left(-i\omega + \frac{1}{\tau}\right)^2 + \left(\frac{qB_0}{m}\right)^2\right\}\vec{v}_\perp(\omega) = \frac{q}{m}\left(-i\omega + \frac{1}{\tau}\right)\vec{E}_\perp + \frac{qB_0}{m}\vec{E}_\perp\times\hat{z}$$

For electrons $q = -e$, $\Omega_c = \frac{eB_0}{m}$

$$\vec{V}_\perp = -\frac{e}{m} \frac{\left(-i\omega + \frac{1}{\tau}\right)\vec{E}_\perp(\omega) + \Omega_c\, \hat{z} \times \vec{E}_\perp(\omega)}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2}$$

$$\hat{z} \times \vec{E}_\perp = \hat{z} \times \left(E_x \hat{x} + E_y \hat{y}\right) = E_x \hat{y} - E_y \hat{x}$$

$$\begin{pmatrix} V_x \\ V_y \end{pmatrix} = -\frac{e}{m} \frac{1}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$$\begin{pmatrix} j_x \\ j_y \end{pmatrix} = -en \begin{pmatrix} V_x \\ V_y \end{pmatrix} = \frac{e^2 n}{m} \frac{1}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$$\Rightarrow \quad j_i = \sigma_{ij} E_j$$

$$\sigma_{ij}^{\perp} = \sigma_{ij} = \frac{e^2 n}{m} \begin{pmatrix} \frac{-i\omega + \frac{1}{\tau}}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} & \frac{-\Omega_c}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \\ \frac{\Omega_c}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} & \frac{-i\omega + \frac{1}{\tau}}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2} \end{pmatrix} ; \quad \sigma = \begin{pmatrix} \sigma_{ij}^{\perp} & 0 \\ 0 & \sigma_{zz} \end{pmatrix}$$

3x3 matrix

whereas along z we only have $\sigma_{zz} = \frac{e^2 n}{m} \frac{1}{-i\omega + \frac{1}{\tau}}$

since Lorentz force does not affect V_z.

Cyclotron resonance: AC field, $\omega \neq 0$

Assume clean conductor so that $\omega \gg \frac{1}{\tau}$

Suppose also $\vec{E}_\perp = \hat{x}\, E_0\, e^{-i\omega t}$ (linearly polarized external field)

$$\Rightarrow \overline{P} = \frac{1}{4}\left(\sigma_{ij} + \sigma_{ji}^*\right) E_i^* E_j \rightarrow \frac{1}{2}\sigma'_{xx}\,|E_0|^2$$

$$\sigma'_{xx} = \frac{e^2 n}{m}\, \mathrm{Re}\left(\frac{-i\omega + \frac{1}{\tau}}{\Omega_c^2 + \left(-i\omega + \frac{1}{\tau}\right)^2}\right) = \frac{e^2 n \tau}{m} \cdot$$

$$\cdot\, \mathrm{Re}\left(\frac{1 - i\omega\tau}{(\Omega_c\tau)^2 + (1 - i\omega\tau)^2}\right)$$

$$\mathrm{Re}\,\frac{1 - i\omega\tau}{(\Omega_c\tau)^2 + 1 - (\omega\tau)^2 - i\,2\omega\tau} =$$

$$= \mathrm{Re}\,\frac{(1 - i\omega\tau)\left(1 + (\Omega_c^2 - \omega^2)\tau^2 + i\,2\omega\tau\right)}{\left(1 + (\Omega_c^2 - \omega^2)\tau^2\right)^2 + 4\omega^2\tau^2}$$

$$= \frac{1 + (\Omega_c^2 - \omega^2)\tau^2 + 2\omega^2\tau^2}{1 + (\Omega_c^2 - \omega^2)^2\tau^4 + 2(\Omega_c^2 - \omega^2)\tau^2 + 4\omega^2\tau^2}$$

$$= \frac{1 + (\Omega_c^2 + \omega^2)\tau^2}{1 + (\Omega_c^2 - \omega^2)^2\tau^4 + 2(\Omega_c^2 + \omega^2)\tau^2}$$

Can neglect 1 relative to $\omega\tau \gg 1$, $\Omega_c\tau \gg 1$

$$\Rightarrow \sigma'_{xx} = \sigma_0 \frac{(\Omega_c^2+\omega^2)\tau^2}{(\Omega_c^2-\omega^2)^2\tau^4 + 2(\Omega_c^2+\omega^2)\tau^2}$$

Note: for $\tau \to \infty$ $\sigma'_{xx} \to \sigma_0 \dfrac{(\Omega_c^2+\omega^2)}{(\Omega_c^2-\omega^2)^2\tau^2} \to 0$

so that absorption is determined by τ.

But this result also has another nice feature $\longrightarrow$ resonance:

for $\Omega_c \approx \omega$ $\quad \sigma'_{xx} = \sigma_0 \dfrac{2\omega^2\tau^2}{4\omega^2\tau^2(\Omega_c-\omega)^2\tau^2 + 4\omega^2\tau^2}$

$\Omega_c^2-\omega^2 = (\Omega_c-\omega)(\Omega_c+\omega)$

$\cong 2\omega(\Omega_c-\omega)$

$$= \frac{1}{2}\sigma_0 \frac{1}{1+(\Omega_c-\omega)^2\tau^2}$$

$$\Rightarrow \overline{P}(\omega) \Rightarrow \frac{1}{4}\frac{\sigma_0}{1+(\omega-\Omega_c)^2\tau^2}|E_0|^2$$

Absorption has maximum at $\omega = \Omega_c$ and is reduced by $\frac{1}{2}$ at $\omega = \Omega_c \pm \frac{1}{\tau}$.

Also: the resonance is <u>chiral</u>

$$\vec{E}_\perp = \hat{x}\, E_0 e^{-i\omega t} \pm i\,\hat{y}\, E_0 e^{-i\omega t}$$

$$\overline{P} = \frac{1}{4}\left(2\sigma'_{xx} + 2\sigma'_{yy}\right)|E_0|^2 + \frac{1}{4}\left(\sigma_{xy} + \sigma^*_{yx}\right) E_0^* \cdot (\pm i E_0)$$

$$+ \frac{1}{4}\left(\sigma_{yx} + \sigma^*_{xy}\right)(\pm i E_0)^* E_0 = \left(\text{use } \sigma_{yx} = -\sigma_{xy}\right)$$

$$= \sigma'_{xx}|E_0|^2 + \frac{1}{4}\left(\sigma_{xy} - \sigma^*_{xy}\right)(\pm i)|E_0|^2$$

$$+ \frac{1}{4}\left(-\sigma_{xy} + \sigma^*_{xy}\right)(\mp i)|E_0|^2 =$$

$$= \left[\sigma'_{xx} + \frac{1}{4}\left(2i\,\sigma''_{xy}(\pm i) - \frac{1}{4}\,2i\,\sigma''_{xy}(\mp i)\right)\right]|E_0|^2$$

$$= \left(\sigma'_{xx} \mp \sigma''_{xy}\right)|E_0|^2.$$

$$\sigma''_{xy} = \mathrm{Im}\,\sigma_0\left(\frac{-\Omega_c\tau}{(\Omega_c\tau)^2 + (1-i\omega\tau)^2}\right) = \sigma_0(-\Omega_c\tau)\cdot$$

$$\cdot\,\mathrm{Im}\left[\frac{1 + (\Omega_c^2 - \omega^2)\tau^2 + i\,2\omega\tau}{\left(1 + (\Omega_c^2-\omega^2)\tau^2\right)^2 + 4\omega^2\tau^2}\right] = \frac{-\sigma_0\,\Omega_c\tau\,(2\omega\tau)}{(\Omega_c^2-\omega^2)^2\tau^4 + 2(\Omega_c^2+\omega^2)\tau^2}$$

$$\Rightarrow \quad \sigma'_{xx} \mp \sigma''_{xy} \sim (\Omega_c^2 + \omega^2)\tau^2 \mp (-2\,\Omega_c\,\omega\,\tau^2)$$

$$= (\Omega_c \cdot \pm\omega)^2\tau^2$$

So that $\overline{P} = \sigma_0 \dfrac{(\Omega_c \pm \omega)^2\tau^2}{4\omega^2\tau^2\left[1 + (\Omega_c - \omega)^2\tau^2\right]}|E_0|^2$

for $\omega \approx \Omega_c$

$$= \frac{1}{4}\sigma_0|E_0|^2\,\frac{\left(1 \pm \frac{\Omega_c}{\omega}\right)^2}{1 + (\Omega_c - \omega)^2\tau^2}$$

The resonance is <u>chiral</u>:
enhanced for $\oplus$ circular polarization and suppressed for $\ominus$.